Self-Operating Charging Network — One-Page Service Agreement

Project Sentinel · Kempower Challenge, LahtiES Energy Tech Hackathon 2026 · Service Provider = equipment supplier & service provider (`contracts.csv`, the party holding 18/29 service-scope contracts) · Client = fleet operator / site owner · Scenario reference day **2026-11-13** · All figures computed from the **synthetic** official dataset; the 2029 customer is fictional.

1 · WHAT WE PROMISE

Commitment	Target	Measured as
SLO-1 Departures	≥ 99.5 % Titanium ≥ 99.9 % Silver ≥ 99.0 %	monthly denominator = all scheduled departures; a departure is met when SoC_{dep} ≥ `required_soc` and delay ≤ 0 min, provided MRDT was met.
SLO-2 Availability	≥ 99.5 %	per-gun minutes available ÷ 720 h/month (per dispenser, not per site).
SLO-3 Off-grid	≥ 72 h, full power	WAN loss from `connectivity_outages.csv`: station holds **1,425 kW DC** (**1,492.15 kW AC ≤ 1,500 kW**, η = 0.955) instead of the 50 kW/dispenser fallback (500 kW).
SLO-4 Grid limit	100 %	`P_AC(t) = P_DC(t)/0.955 ≤ P_limit(t)` evaluated on the time-varying limit; **0** official over-limit rows in **134,400** intervals is the reference.
SLO-5 Latency	< 10 ms decide 85 ms module 260 ms relay	software decision, intra-cabinet 25 kW module isolation, and module relay break-before-make. The macro inter-cabinet re-route is 2.63 s — a different path, not a faster promise.

Carve-outs from the measurement window (they suspend the clock, they do not lower the bar): an *unannounced* medium-voltage grid outage or DSO hard curtailment at a site with no storage — `bess_kwh = 0` at all five close-up sites — is excluded from SLO-1 and SLO-2 from the instant it occurs; and any arrival whose dwell time < MRDT is *best-effort*, not a breach. An *announced* curtailment such as G1 (notified 14:00, active 16:00–19:00) stays inside the promise, because the system has time to re-plan.

Outside SLO scope by design: weather, vehicle-side faults, driver arrival lateness below MRDT, and the travel time of a physical module replacement (see §3).

2 · WHAT THE SYSTEM DECIDES ALONE

- **How much power each gun gets** — intra-cabinet DPR across 25 kW modules, continuous allocation, strict 1:N wiring as recorded in `chargers.csv`.
- **When and how fast each vehicle charges** — 15-minute rolling MILP (SciPy HiGHS) on `prices.csv` + `fleet_schedule.csv`, departure deadline weighted above price, so a feasible vehicle is never dropped to save money.
- **Who gets curtailed when the grid is cut** — e.g. S2 under G1 (notified 14:00, active **16:00–19:00** at 400 kW): overnight trucks T05–T08 throttled with continuous setpoints, **283.50 kWh** shifted to afternoon trucks T01–T04 → **4** → **0** missed.
- **When to isolate, reroute and re-level** — thermal precursor (S1: 59.3 °C / 702 rpm at 22:31, **199 min** lead), N-1 degradation to **720 kW** with 0 kW self-harm, offline islanding.
- **What a station's configured limit actually is** — three-way reconciliation of `network_sites.csv` / `power_cabinets.csv` / `site_config.csv` (S5: 600 → 900 kW, +750 kWh).
- **Whether a claim is payable** — MRDT gate evaluated per session before any credit is issued.

$$MRDT_j = \Delta SoC \cdot E_{batt} \div \left(\min(P_{charger}, P_{veh}) \cdot \eta_{sys} \right) \times 60 + 15\ \text{min}$$ → worked example **57.11 min** (`verify_all.py`).

Human decisions kept by the Client: site acceptance, tariff/contract parameters, any hardware purchase (§3), and physical repair dispatch. The system never overrides a local safety interlock.

3 · ACCOUNTABILITY WHEN IT DECIDES WRONG

Event	Payable by the Service Provider
Departure below required SoC / late	350 € Titanium · 250 € · 150 € per trip
Delay beyond the tier threshold	7.50 / 5.00 / 3.00 €/min (from 10 / 15 / 20 min)
Energy shortfall	2.00 / 1.50 / 1.00 €/kWh (+ emergency hire in Titanium)
Availability below target	−2 % of the monthly fee per 0.1 pp shortfall, capped at 90 %; 0 % fee if < 95 %

- **Hard cap:** aggregate monthly liability = **150 %** of the monthly base fee. Invoices are non-negative; the Client can never be billed back.
- **Client-side guard:** if dwell time < MRDT, the session is *best-effort* and no liquidated damages accrue — the gate that stops an arrival-10-minutes-late-with-200-kWh-to-go claim.
- **What we do not take liability for:** a cabinet that needs a human on site. `faults.csv` clears **8** of **19** faults with `cleared_by = technician`; those **73** dispatches = **22,906 min** = **5.09 FTE** are **retained, not sold**. Same for **3.22 FTE** of customer contact.
- **Zero-miss boundary, stated:** on 2026-11-13 our zero-CAPEX software takes S1 **4** → **3** missed (**398.5** → **48.8 kWh, −87.8 %**) with **12** → **0** manual replugs. Reaching **0** needs an inter-cabinet busbar tie that is absent from `chargers.csv` and has **no price** in `equipment_costs.csv` — we hand that number back rather than invent it. Buying 12 dispensers alone (**324,000 €**) yields **8** missed, i.e. worse.

Why a client signs this: under the current contracts three of five close-up sites already breach their own uptime floor — recorded **S1 94.1 %**, **S3 90.0 %**, **S5 87.5 %** vs a **97 %** commitment (`C1`), at **150 €** per undercharged departure = **1,350 €** on the scenario day against a **263 €** daily fee. The penalty is already flowing.

Basis of every number: unmodified official CSVs — 45 files / 240,635 rows; recomputed by `python verify_all.py` (< 1 s, exit 0) and `tools/export_deck_ledger.py`. Labour baseline **8,852** ops / **93,412** min / **20.76 FTE** at 37.5 h·wk and 45 €/h, of which **12.44 FTE (59.95 %)** is software-addressable; grid-limit capping is done by a person today — **684** manual load adjustments / **10,260** min, **100 %** of them at sites breaching their limit (Spearman **0.802**). Assumptions and their error direction: `docs/assumptions_and_additions.md`. This one-pager is the deliverable; full derivations, tier tables and the 2029 customer story: `SLA_Specification_Annex_Full.pdf`.

Service Provider — name / signature / date

Client — name / signature / date